

Endian packing

The same integer can sit in memory two ways

Unpack, pack, swap

- Unpack splits a word into bytes in address order for the chosen mode
- Pack builds the word from bytes at plus zero through plus three
- Byte-swap reverses the four bytes in the word
- Compare LE versus BE when you want the side-by-side picture
- BE puts one-two first

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / Endian

INTERACTIVE TOOL

Endian packing

See **big-endian** vs **little-endian** byte order — pack/unpack a 32-bit word and watch memory lanes.

Starter example: Word 0x12345678. Little-endian memory is 78 56 34 12; big-endian is 12 34 56 78. Flip the mode and pack/unpack.

Load starter example

Challenges 0 / 22 cleared

Quiz: LE: Little-endian stores at lowest address the? Answer: lsb

Answer

Show hint

Check

Next

Idle

Quiz: LE

Quiz: BE

Quiz: network

Quiz: value

Starter word

Unpack LE

Unpack BE

Compare

Byte-swap

Pack LE

Pack BE

Quiz: LE first

Quiz: BE first

Set DEADBEEF

Preset

Mode switch

Quiz: host

Round-trip

Swap twice

Quiz: dump

Edit byte

Full endian

Core ideas

Little-endian

Lowest address stores the **LSB** of the word.

Big-endian

Lowest address stores the **MSB** — common on the wire.

Endian lab starter

Workbook practice

- In the workbook track, take word one-two-three-four-five-six-seven-eight
- Write LE and BE byte order at addresses plus zero through plus three
- Pack LE bytes A-A B-B C-C D-D and check you get D-D C-C B-B A-A
- Note one pitfall: assuming host endian always matches the wire or the spec

Pitfalls to watch

- Do not confuse byte order with bit order inside a byte
- Mixing LE and BE across a bus without conversion corrupts multi-byte fields
- And remember: the browser lab is literacy
- Real SoCs still need explicit endian policy on DMA, registers, and network frames

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, unpack one word both ways and pack one byte sequence
- When you are ready, take the short quiz, then continue to truth tables